

Unifying Special Relativity and Quantum Mechanics

Lecture 7 — Relativistic Causality and Quantum Fields: Microcausality, Transformation Laws, and the Particle–Field Connection

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as Lectures 1–6: everything in the normal text and in the yellow boxes is a faithful transcription of the handwritten notes (yellow boxes correspond to the results that were highlighted in the original, whether in yellow, pink/red, or framed in color); everything inside a blue box like this one is extra material that I added. The figure is a redrawn version of the hand sketch.

Pagination of this scan. The scan continues the triple-starred sequence opened by the “From particles to fields” chapter at the end of Lecture 6: its pages carry the circled numbers 9–26, i.e. they are [p. 9***]–[p. 26***]. One anomaly: the sheets numbered 13 and 14 appear in the scan in the order 14, then 13, and the content flows continuously in exactly that order (the infinitesimal-translation expansion at the bottom of the sheet marked 14 is completed by “from the comparison, we get...” at the top of the sheet marked 13) — evidently a slip in the handwritten numbering. The markers below follow the scan/content order: ..., [p. 12***], [p. 14***], [p. 13***], [p. 15***], ...

Where we left off (recap of Lecture 6). The multi-particle (Fock) framework was set up: creation/annihilation operators with $[a, a^\dagger]_\mp = (2\pi)^3 \delta(\vec{p} - \vec{q}) \delta_{\lambda\lambda'} \delta_{mm'}$ (commutators for integer spin, anticommutators for half-integer spin), the free Hamiltonian H_0 counting single-particle energies, and the Poincaré transformation laws

$$U(\Lambda, a) a^\dagger(M, \vec{p}, s, \sigma) U(\Lambda, a)^{-1} = e^{ia \cdot p_\Lambda} \left(\frac{E_{p_\Lambda}}{E_p} \right)^{1/2} \sum_{\sigma'} [D_s(W_L(\Lambda, p))]_{\sigma'\sigma} a^\dagger(M, \vec{p}_\Lambda, s, \sigma'),$$

and its Hermitian conjugate for a (with D_s^*), where $W_L(\Lambda, p) = L(p_\Lambda)^{-1} \Lambda L(p)$ is the Wigner rotation of the spin-polarization basis. The lecture ended announcing “Point 2” of the new framework: localize *observations* rather than states, implementing *relativistic causality*. That sentence is completed by the first yellow box below.

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1	Relativistic causality and observables [p. 9***]	

$$[O_1(x), O_2(y)] = 0 \quad \text{if } (x - y)^2 < 0$$

$O_1(x)$: observable measured at space-time position “ x ”;

$O_2(y)$: observable measured at space-time position “ y ”;

$\triangle$ they must commute on space-like separated intervals.

“Observable” means Hermitian operator-valued function of space-time ($\triangle$ must be Hermitian to be an observable).

Hermitian operator-valued functions of space-time that commute at space-like separations are expressed as products of one or more elementary building blocks called quantum fields:

$$\Phi^A(x) \equiv \Phi^A(t, \vec{x})$$

such that:

1) Individually, it is not necessary for a quantum field to be Hermitian

$\hookrightarrow$ what matters is that the observables they define by combining are.

2 Microcausality: fields either commute or anticommute [p. 10***]

2) At space-like separated points, quantum fields either commute or anticommute:

$$[\Phi^A(x), \Psi^B(y)]_{\pm} = 0, \quad (x - y)^2 < 0 \quad \text{MICROCAUSALITY CONDITION}$$

We are saying that, at the level of quantum fields, there are two ways of implementing relativistic causality at the level of observable operators,

$$[O_1(x), O_2(y)] = 0, \quad (x - y)^2 < 0 :$$

Simplest possibility: consider commuting

In this case, any observable made out of a product of fields will commute on $(x - y)^2 < 0$:

$$[\Phi^A(x), \Psi^B(y)]_{\mp} = 0 \text{ for } (x - y)^2 < 0.$$

it is still possible to have observables that commute if they are made of an *even* number of anti-commuting quantum fields:

$$[AB, C] = A\{B, C\} - \{A, C\}B.$$

Added explanation (not in the original notes)

Why causality is attached to observables. With the mostly-minus metric of Lecture 2, $(x - y)^2 < 0$ means the two events are space-like separated: no signal can connect them. If two observables at such points failed to commute, measuring one would disturb the statistics of the other — an instantaneous influence. Demanding $[O_1(x), O_2(y)] = 0$ is therefore the operator statement of relativistic causality (“no superluminal signalling”), and it replaces the failed attempts of Lecture 6 to localize *states*. Note the structure of the argument: causality constrains only the Hermitian *composites*, so the elementary building blocks have exactly two options, closed under the identities displayed above (the two mixed-bracket identities are the $(\star)$ identity of Lecture 6, [p. 5***], written for the two sign choices). The even-number rule in the anticommuting case is why fermionic observables — energy density, currents, ... — are always *bilinear* in the fermion fields: a single anticommuting field can never be an observable. Which option pairs with which spin is precisely the spin-statistics correspondence stated (as an input) in Lecture 6; the computation on [p. 25***] below is the first step of the argument that *forces* the pairing.

3 Non-trivial transformation properties under Poincaré [p. 11***]

3) Quantum fields have non-trivial transformation properties under Poincaré.

Obs. In QM a transformation acting on the states

$$|\psi\rangle \longrightarrow |\psi'\rangle = U|\psi\rangle$$

can be also seen as a transformation acting on the operators:

$$\langle\varphi|O|\psi\rangle \longrightarrow \langle\varphi|U^\dagger O U|\psi\rangle \equiv \langle\varphi|O'|\psi\rangle, \quad \text{that is} \quad \boxed{O \rightarrow O' = U^\dagger O U}$$

Here the logic is the same of this formula:

$$\Phi^A(x) \xrightarrow{(\Lambda, a)} \Phi'^A(x) = U^\dagger(\Lambda, a) \Phi^A(x) U(\Lambda, a) = (D(\Lambda))^A_B \Phi^B(\Lambda^{-1}x - \Lambda^{-1}a)$$

$U(\Lambda, a)$: unitary operators that implement the action of Poincaré transformations on the states;

$D(\Lambda)$: finite-dimensional representation of the Lorentz group;

the index “ A ” runs over a discrete and finite set of values. We should think about $\Phi^A(x)$ as the component of an object that transforms non-trivially under Lorentz.

Added explanation (not in the original notes)

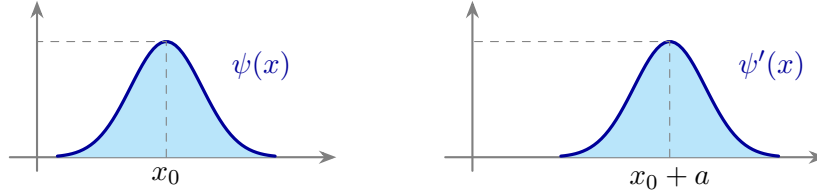
A point worth dwelling on: $D(\Lambda)$ is finite-dimensional, hence *not* unitary. The course proved (Lecture 2, [p. 43]-box, and throughout) that the physically interesting *unitary* representations of the non-compact Lorentz/Poincaré group are infinite-dimensional — those are the $U(\Lambda, a)$ acting on the Hilbert space of states. The matrices $D(\Lambda)$ appearing here act instead on the handful of *components* of the field, so they form a finite-dimensional representation, which for a non-compact group is necessarily non-unitary (for boosts, $D(\Lambda)^\dagger \neq D(\Lambda)^{-1}$). There is no contradiction: unitarity is required where probabilities live (the states), not on the bookkeeping index A . The finite-dimensional representations of the Lorentz group (scalar, Weyl/Dirac spinor, vector, ...) will presumably be classified later in the course; the two-level structure — U on states, D on components — is the defining tension of quantum fields, and the consistency conditions of [p. 22***] below are exactly the bridge between the two levels.

4 Comment: the two levels of the transformation [p. 12***]

$$\Phi^A(x) \xrightarrow{\Lambda, a} \Phi'^A(x) = (D(\Lambda))^A_B \Phi^B(\Lambda^{-1}x - \Lambda^{-1}a) \quad (\star)$$

$\rightsquigarrow$ Lorentz transformations act at two levels: they reshuffle the components of the fields and also transform the space-time dependence.

About this last part, consider the following example.



Consider a translation acting on a function $\psi(x)$. The transformed function is $\psi'(x)$. We then have

$$\psi'(x_0 + a) = \psi(x_0).$$

If we set $x_0 + a \equiv x \rightarrow x_0 = x - a$ and

$$\psi'(x) = \psi(x - a)$$

$\rightsquigarrow$ the value of the new function at “ x ” is equal to the value of the old function evaluated at the “inverse transformed point” $x - a$.

This argument explains the functional dependence in $(\star)$.

[p. 14***] In the transformation law

$$\Phi^A(x) \rightarrow \Phi'^A(x) = (D(\Lambda))^A_B \Phi^B(\Lambda^{-1}x - \Lambda^{-1}a),$$

this is indeed the “inverse transformed point” of $x \rightarrow \Lambda x + a$, which is $x \rightarrow \Lambda^{-1}x - \Lambda^{-1}a$. In addition, we are saying that the quantum fields $\Phi^A(x)$ non-trivially transform under Lorentz.

It is interesting to ask what physical properties are delineated by the non-trivial transformation under Poincaré. We investigate separately space-time translations and Lorentz transformations.

5 Space-time translations: $[P^\mu, \Phi^A(x)]$ [p. 14***–13***]

We consider infinitesimal transformations:

$$U(\mathbb{1}, \varepsilon)^\dagger \Phi^A(x) U(\mathbb{1}, \varepsilon) = \Phi^A(x - \varepsilon), \quad U(\mathbb{1}, \varepsilon) = \mathbb{1} + i\varepsilon_\mu P^\mu,$$

$$\Phi^A(x - \varepsilon) = \Phi^A(x) - \varepsilon_\mu \partial^\mu \Phi^A(x) + O(\varepsilon^2).$$

Expanding the left-hand side:

$$(\mathbb{1} - i\varepsilon_\mu P^\mu) \Phi^A(x) (\mathbb{1} + i\varepsilon_\mu P^\mu) = \Phi^A(x) - \varepsilon_\mu \partial^\mu \Phi^A(x) + O(\varepsilon^2)$$

$$\cancel{\Phi^A(x)} - i\varepsilon_\mu [P^\mu, \Phi^A(x)] + O(\varepsilon^2) = \cancel{\Phi^A(x)} - \varepsilon_\mu \partial^\mu \Phi^A(x) + O(\varepsilon^2).$$

[p. 13***] From the comparison, we get

$$[P^\mu, \Phi^A(x)] = -i \partial^\mu \Phi^A(x)$$

where P^μ is the Hermitian operator, generator of space-time translations in the unitary representation acting on states. In components:

$$\mu = 0 : \quad [P^0, \Phi^A(x)] = -i \partial_t \Phi^A(x) \quad \text{describes the dynamical evolution of the field;}$$

$$\mu = 1, 2, 3 : \quad [\vec{P}, \Phi^A(x)] = i \vec{\nabla} \Phi^A(x).$$

Added explanation (not in the original notes)

This is the Heisenberg equation of motion. Writing the $\mu = 0$ component as $i \partial_t \Phi^A = [\Phi^A, H]$ (with $H = P^0$) shows that the abstract group-theoretic statement “ P^0 generates time translations” is the Heisenberg equation of quantum mechanics: dynamics has been absorbed into the representation theory. The finite version, obtained by exponentiating, is $\Phi^A(x) = e^{iP \cdot x} \Phi^A(0) e^{-iP \cdot x}$: the entire spacetime dependence of a free field is generated from its value at one point by the momentum operators. (This identity is used implicitly on [p. 23***] below, where pure translations fix the coefficient functions to be plane waves.) On the sign conventions: with $U(\mathbb{1}, a) = e^{ia_\mu P^\mu}$ as in Lecture 6 and $\partial^\mu = g^{\mu\nu} \partial_\nu$, the spatial components carry the opposite sign, whence the $+i \vec{\nabla}$ in the second box. Note also the small notational drift in the original: from this page on the component index is written interchangeably as Φ^a or Φ^A ; transcribed uniformly as Φ^A .

6 Lorentz transformations: $[J^{\mu\nu}, \Phi^A(x)]$ [p. 13***, 15***]

$$U^\dagger(\Lambda) \Phi^A(x) U(\Lambda) = D(\Lambda)^A_B \Phi^B(\Lambda^{-1}x), \quad \Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad (\Lambda^{-1})^\mu{}_\nu = \delta^\mu{}_\nu - \omega^\mu{}_\nu,$$

$$U(\Lambda(\omega)) = \mathbb{1} - \frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}, \quad D(\Lambda(\omega)) = \mathbb{1} - \frac{i}{2} \omega_{\mu\nu} J_S^{\mu\nu},$$

where $J^{\mu\nu}$ are the Lorentz generators in the unitary irrep acting on states, and $J_S^{\mu\nu}$ are the Lorentz generators in the representation $D(\Lambda)$ according to which the fields transform.

[p. 15***] Expanding to first order:

$$\begin{aligned} \left(\mathbb{1} + \frac{i}{2} \omega_{\mu\nu} J^{\mu\nu} \right) \Phi^A(x) \left(\mathbb{1} - \frac{i}{2} \omega_{\mu\nu} J^{\mu\nu} \right) &= \left(\delta^A_B - \frac{i}{2} \omega_{\mu\nu} (J_S^{\mu\nu})^A_B \right) \Phi^B(x - \omega x) \\ \Phi^A(x) + \frac{i}{2} \omega_{\mu\nu} [J^{\mu\nu}, \Phi^A(x)] &= \left(\delta^A_B - \frac{i}{2} \omega_{\mu\nu} (J_S^{\mu\nu})^A_B \right) \Phi^B(x - \omega x) \end{aligned} \quad (\otimes)$$

Focus on this term:

$$\begin{aligned} \Phi^B(x - \omega x) &= \Phi^B(x) - (\omega x)^\mu \partial_\mu \Phi^B(x) = \Phi^B(x) - \omega^\mu{}_\nu x^\nu \partial_\mu \Phi^B(x) \\ &= \Phi^B(x) - \omega_{\mu\nu} x^\nu \partial^\mu \Phi^B(x) = \Phi^B(x) - \frac{\omega_{\mu\nu}}{2} (x^\nu \partial^\mu - x^\mu \partial^\nu) \Phi^B(x) \\ &= \Phi^B(x) - \frac{i \omega_{\mu\nu}}{2} i (x^\mu \partial^\nu - x^\nu \partial^\mu) \Phi^B(x). \end{aligned}$$

Going back to $\otimes$, at the first order in $\omega_{\mu\nu}$, we find

$$\cancel{\Phi^A(x)} + \frac{i}{2} \omega_{\mu\nu} [J^{\mu\nu}, \Phi^A(x)] = \cancel{\Phi^A(x)} - \frac{i}{2} \omega_{\mu\nu} (J_S^{\mu\nu})^A_B \Phi^B(x) - \frac{i}{2} \omega_{\mu\nu} i (x^\mu \partial^\nu - x^\nu \partial^\mu) \delta^A_B \Phi^B(x),$$

that is [p. 16***]

$$[J^{\mu\nu}, \Phi^A(x)] = - \left[(J_S^{\mu\nu})^A_B + i (x^\mu \partial^\nu - x^\nu \partial^\mu) \delta^A_B \right] \Phi^B(x)$$

Added explanation (not in the original notes)

Bookkeeping of the antisymmetrization. The only nontrivial step in the chain is $\omega_{\mu\nu} x^\nu \partial^\mu = \frac{\omega_{\mu\nu}}{2} (x^\nu \partial^\mu - x^\mu \partial^\nu)$: since $\omega_{\mu\nu}$ is antisymmetric, only the antisymmetric part of $x^\nu \partial^\mu$ in $(\mu\nu)$ survives, and writing it out explicitly costs the factor $\frac{1}{2}$ (relabel $\mu \leftrightarrow \nu$ in half of the term) — the same double-counting mechanics as in the tensor packaging of Lecture 2, [p. 44]. The final rewriting inserts $1 = -i \cdot i$ so that the differential operator appears as $L^{\mu\nu} \equiv i(x^\mu \partial^\nu - x^\nu \partial^\mu)$: a *Hermitian* differential operator (on the invariant measure) with exactly the commutation relations of the Lorentz algebra — it is the infinite-dimensional “orbital” representation of $J^{\mu\nu}$ carried by the argument x , sitting alongside the finite-dimensional “spin” representation $J_S^{\mu\nu}$ carried by the index A . The boxed formula says: the total generator splits as spin + orbital, mirroring at the field level the two-level structure of the transformation law $(\star)$. The overall minus sign is a consequence of the $O \rightarrow U^\dagger O U$ (operator-transformation) convention chosen on [p. 11***].

7 Rotations: orbital angular momentum and spin [p. 16***–17***]

Focus on rotations, that is, let’s compute

$$[J^i, \Phi^A(x)] = \frac{1}{2} \varepsilon_{ijk} [J^{jk}, \Phi^A(x)] \quad (\text{we read this commutator from the previous result}).$$

$$\begin{aligned} [J^i, \Phi^A(x)] &= - \left(\frac{1}{2} \varepsilon_{ijk} (J_S^{jk})^A_B + \frac{i}{2} \varepsilon_{ijk} (x^j \partial^k - x^k \partial^j) \delta^A_B \right) \Phi^B(x) \\ &= - \left(\underbrace{\frac{1}{2} \varepsilon_{ijk} (J_S^{jk})^A_B}_{\text{this is } J_S^i} + \frac{i}{2} \varepsilon_{ijk} (-x^j \partial_k + x^k \partial_j) \delta^A_B \right) \Phi^B(x) \end{aligned}$$

(J_S^i : the generators of rotation in the representation according to which fields transform)

$$\begin{aligned} &= -\left((J_S^i)^A{}_B + i\varepsilon_{ijk}(-x^j\partial_k)\delta^A{}_B\right)\Phi^B(x) \\ &= -\left((J_S^i)^A{}_B + \varepsilon_{ijk}x^j(-i\partial_k)\delta^A{}_B\right)\Phi^B(x) = -\left((J_S^i)^A{}_B + (\vec{x} \times (-i\vec{\nabla}))^i\delta^A{}_B\right)\Phi^B(x). \end{aligned}$$

In vector form: [p. 17***]

$$[\vec{J}, \Phi^A(x)] = -\left((\vec{J}_S)^A{}_B + \vec{x} \times (-i\vec{\nabla})\delta^A{}_B\right)\Phi^B(x)$$

$\vec{J}$: Hermitian operator, generator of rotations in the representation acting on the states; it describes the total angular momentum.

$\rightsquigarrow$ the separation between orbital angular momentum and spin is evident:

■ $\vec{L} = \vec{x} \times (-i\vec{\nabla})$ is the *orbital angular momentum*, and it originates from the space-time dependence of the field;

■ $\vec{J}_S$ is the *spin*.

If we consider a field that transforms non-trivially under Lorentz, it means that it carries some spin.

Added explanation (not in the original notes)

The punchline, in familiar terms. The formula is the field-theoretic incarnation of $\vec{J} = \vec{L} + \vec{S}$ from ordinary quantum mechanics — but with the logic reversed. In QM one *postulates* a spin degree of freedom and adds it to $\vec{L}$; here both pieces *emerge* from the transformation law ($\star$): the orbital part from the reshuffling of the argument x (it is the same $\vec{x} \times (-i\vec{\nabla})$ familiar from wave mechanics, with $-i\vec{\nabla} \leftrightarrow \hat{p}$), and the spin part from the reshuffling of the components by $D(\Lambda)$. A field transforming in the trivial representation ($D = \mathbb{1}$, a single component: a *scalar field*) has $\vec{J}_S = 0$ and carries pure orbital angular momentum — it will describe spin-0 particles. The steps $\partial^k = -\partial_k$ (spatial index raised with the mostly-minus metric) and the ε -relabeling $\varepsilon_{ijk}x^k\partial_j = -\varepsilon_{ijk}x^j\partial_k$ are the only mechanics in the chain; $\frac{1}{2}\varepsilon_{ijk}J_S^{jk} = J_S^i$ is the by-now-standard packaging of Lecture 2, [p. 44].

8 The inverse form of the transformation law [p. 17***–18***]

Comment. The transformation property can be rewritten as follows. From

$$U(\Lambda, a)^\dagger \Phi^A(x) U(\Lambda, a) = D(\Lambda)^A{}_B \Phi^B(\Lambda^{-1}x - \Lambda^{-1}a),$$

since U is unitary, $U^\dagger = U^{-1}$ and, using the properties of representations,

$$U(\Lambda, a)^\dagger = U(\Lambda, a)^{-1} = U(\Lambda^{-1}, -\Lambda^{-1}a),$$

so we have [p. 18***]

$$U(\Lambda^{-1}, -\Lambda^{-1}a) \Phi^A(x) U(\Lambda, a) = D(\Lambda)^A{}_B \Phi^B(\Lambda^{-1}x - \Lambda^{-1}a).$$

We now set

$$\begin{aligned} \tilde{\Lambda} &\equiv \Lambda^{-1}; & -\Lambda^{-1}a &\equiv b \rightsquigarrow a = -\Lambda b = -\tilde{\Lambda}^{-1}b, \\ U(\tilde{\Lambda}, b) \Phi^A(x) U(\tilde{\Lambda}^{-1}, -\tilde{\Lambda}^{-1}b) &= D(\tilde{\Lambda}^{-1})^A{}_B \Phi^B(\tilde{\Lambda}x + b). \end{aligned}$$

Since $\tilde{\Lambda}$ is generic, we rename it Λ . We can, therefore, equally well write

$$U(\Lambda, a) \Phi^A(x) U(\Lambda, a)^\dagger = D(\Lambda^{-1})^A{}_B \Phi^B(\Lambda x + a)$$

9 Decomposition in creation and annihilation operators [p. 18***–19***]

4) Decomposition in terms of creation and annihilation operators.

Quantum fields are operators acting on states. We already defined a class of such operators: creation and annihilation operators. We thus make the simplest assumption: *quantum fields are linear combinations of creation and annihilation operators*. We write [p. 19***]

$$\begin{aligned} \Phi_+^A(x) &= \sum_\sigma \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} u^A(m, \vec{p}, \sigma, x) a(m, \vec{p}, \sigma) \\ U(\Lambda, a) \Phi_+^A(x) U(\Lambda, a)^\dagger &= (D(\Lambda^{-1}))^A{}_B \Phi_+^B(\Lambda x + a) \end{aligned}$$

$$\begin{aligned} \Phi_-^A(x) &= \sum_\sigma \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} v^A(m, \vec{p}, \sigma, x) a^\dagger(m, \vec{p}, \sigma) \\ U(\Lambda, a) \Phi_-^A(x) U(\Lambda, a)^\dagger &= (D(\Lambda^{-1}))^A{}_B \Phi_-^B(\Lambda x + a) \end{aligned}$$

Comment:

$$E_p = (|\vec{p}|^2 + m^2)^{1/2} \text{ massive case,} \quad E_p = |\vec{p}| \text{ massless case;}$$

the coefficient function u^A (as well as v^A) carries a Lorentz index;

“ m ” is fixed, depends on the kind of particles (mass, spin) we want to describe.

[p. 20***] “ m ” identifies the irrep of Poincaré. The quantum fields Φ_+ and Φ_- sum over all possible spin polarizations and integrate over all possible momenta since *we want to create or destroy any allowed particle state in the irrep*.

Added explanation (not in the original notes)

Reading the ansatz. Φ_+ (built from a ’s) can only *destroy* particles of the irrep m ; Φ_- (built from $a^\dagger$ ’s) can only *create* them — the subscripts anticipate the standard names “positive-” and “negative-frequency” parts, which become literal once [p. 23***] fixes the x -dependence to $e^{-ip \cdot x}$ and $e^{+ip \cdot x}$ respectively. Both boxes *impose* that the field transform covariantly, with the same finite-dimensional D : that requirement is not automatic — the $a^\dagger$ ’s transform with the infinite-dimensional unitary machinery of Lecture 6 (Wigner rotations, energy ratios), so demanding that the linear combination transform with a finite-dimensional $D(\Lambda^{-1})$ on the index A is a strong constraint on the coefficient functions. Extracting that constraint is the business of the next section, and its solutions u^A, v^A (polarization spinors/vectors/...) are what textbooks tabulate for each spin. Note also the measure: the combination $\frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}}$ times the $\sqrt{2E}$ -normalized operators is engineered so that Lorentz invariance can be made manifest by multiplying and dividing by $\sqrt{E_p}$ (as done on [p. 20***] below), reassembling the invariant measure $d^3\vec{p}/E_p$ of Lecture 5.

10 The particle–field connection: massive case [p. 20***–22***]

The coefficient functions $u^A(m, \vec{p}, \sigma, x)$ and $v^A(m, \vec{p}, \sigma, x)$ are not arbitrary but must satisfy a consistency condition.

► We consider the case of $\Phi_-^A(x)$, and elaborate as follows.

$$\begin{aligned} \underbrace{U(\Lambda, a) \Phi_-^A(x) U(\Lambda, a)^{-1}}_{\circledast} &= D(\Lambda^{-1})^A{}_B \Phi_-^B(\Lambda x + a) \\ &= D(\Lambda^{-1})^A{}_B \sum_{\sigma} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} v^B(m, \vec{p}, \sigma, \Lambda x + a) a^{\dagger}(m, \vec{p}, \sigma) \\ &= D(\Lambda^{-1})^A{}_B \sum_{\sigma} \int \frac{d^3 \vec{p} \sqrt{E_p}}{(2\pi)^3 \sqrt{2} E_p} v^B(m, \vec{p}, \sigma, \Lambda x + a) a^{\dagger}(m, \vec{p}, \sigma). \end{aligned}$$

We change variable $p \rightarrow \Lambda p$, and use the fact that $d^3 \vec{p}/E_p$ is Lorentz-invariant: [p. 21***]

$$\begin{aligned} &= D(\Lambda^{-1})^A{}_B \sum_{\sigma} \int \frac{d^3 \vec{p} \sqrt{E_{p\Lambda}}}{(2\pi)^3 \sqrt{2} E_p} v^B(m, \vec{p}_{\Lambda}, \sigma, \Lambda x + a) a^{\dagger}(m, \vec{p}_{\Lambda}, \sigma) \\ &= D(\Lambda^{-1})^A{}_B \sum_{\sigma} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left(\frac{E_{p\Lambda}}{E_p} \right)^{1/2} v^B(m, \vec{p}_{\Lambda}, \sigma, \Lambda x + a) a^{\dagger}(m, \vec{p}_{\Lambda}, \sigma) \end{aligned} \quad (1)$$

We also elaborate on the left-hand side $\circledast$ as follows:

$$\begin{aligned} U(\Lambda, a) \Phi_-^A(x) U(\Lambda, a)^{-1} &= U(\Lambda, a) \left[\sum_{\sigma} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} v^A(m, \vec{p}, \sigma, x) a^{\dagger}(m, \vec{p}, \sigma) \right] U(\Lambda, a)^{-1} \\ &= \sum_{\sigma} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} v^A(m, \vec{p}, \sigma, x) \underbrace{\left[U(\Lambda, a) a^{\dagger}(m, \vec{p}, \sigma) U(\Lambda, a)^{-1} \right]}_{\text{transformation properties of creation operators}} \\ &= \sum_{\sigma} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} v^A(m, \vec{p}, \sigma, x) \times \\ &\quad \times \left[e^{ia \cdot p_{\Lambda}} \left(\frac{E_{p\Lambda}}{E_p} \right)^{1/2} \sum_{\sigma'} (D_s(W_L(\Lambda, p)))_{\sigma'\sigma} a^{\dagger}(m, \vec{p}_{\Lambda}, \sigma') \right] \end{aligned}$$

We relabel $\sigma \leftrightarrow \sigma'$:

$$= \sum_{\sigma, \sigma'} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} v^A(m, \vec{p}, \sigma', x) e^{ia \cdot p_{\Lambda}} \left(\frac{E_{p\Lambda}}{E_p} \right)^{1/2} (D_s(W_L(\Lambda, p)))_{\sigma\sigma'} a^{\dagger}(m, \vec{p}_{\Lambda}, \sigma) \quad (2)$$

From the comparison between (1) and (2) we get [p. 22***]

$$D(\Lambda^{-1})^A{}_B v^B(m, \vec{p}_{\Lambda}, \sigma, \Lambda x + a) = \sum_{\sigma'} v^A(m, \vec{p}, \sigma', x) e^{ia \cdot p_{\Lambda}} (D_s(W_L(\Lambda, p)))_{\sigma\sigma'}$$

► There is an equivalent equation for u^A . The difference comes only from the slightly different transformation properties under Poincaré of the annihilation operators. We have

$$D(\Lambda^{-1})^A{}_B u^B(m, \vec{p}_{\Lambda}, \sigma, \Lambda x + a) = \sum_{\sigma'} u^A(m, \vec{p}, \sigma', x) e^{-ia \cdot p_{\Lambda}} (D_s(W_L(\Lambda, p)))_{\sigma\sigma'}^*.$$

These matrices are unitary, so that we can write

$$D_s(W_L)^\dagger = D_s(W_L)^{-1} \Rightarrow D_s(W_L)^{T*} = D_s(W_L^{-1})$$

and, in components,

$$(D_s(W_L^{-1}))_{\sigma\sigma'} = (D_s(W_L)^{*T})_{\sigma\sigma'} = (D_s(W_L))_{\sigma'\sigma}^*,$$

so that

$$D(\Lambda^{-1})^A_B u^B(m, \vec{p}_\Lambda, \sigma, \Lambda x + a) = \sum_{\sigma'} u^A(m, \vec{p}, \sigma', x) e^{-ia \cdot p_\Lambda} (D_s(W_L(\Lambda, p)^{-1}))_{\sigma'\sigma}$$

We look at what happens to these equations if we restrict to pure space-time translations.

Added explanation (not in the original notes)

Anatomy of the consistency condition. The derivation is once again a *matching* argument (the third of the course, after the operator identities of Lecture 1, [p. 20], and the $a^\dagger$ transformation law of Lecture 6, [p. 7***]): the same object $\otimes$ is computed once using the *postulated field* transformation (D acting on the index, argument shifted to $\Lambda x + a$) and once using the *known operator* transformation of $a^\dagger$ (Wigner rotation, phase, energy ratio); equating the coefficients of the linearly independent operators $a^\dagger(m, \vec{p}_\Lambda, \sigma)$ gives the boxed condition. Two technical points worth flagging: (i) the change of variables $\vec{p} \rightarrow \vec{p}_\Lambda$ in (1) is exactly where the Lorentz-invariant measure $d^3\vec{p}/E_p$ (Lecture 5) earns its keep — and the leftover ratio $(E_{p_\Lambda}/E_p)^{1/2}$ from the measure precisely cancels the one produced by the $a^\dagger$ transformation law, as the final boxed equations (which contain *no* energy ratios) confirm; (ii) the relabeling $\sigma \leftrightarrow \sigma'$ in (2) is legitimate because both are dummy summation indices — it just aligns the free index with the one in (1). The unitarity manipulation $D_s(W_L)_{\sigma\sigma'}^* = D_s(W_L^{-1})_{\sigma'\sigma}$ trades the complex-conjugated matrix for the matrix of the inverse rotation, putting the u -equation in the same algebraic shape as the v -equation — the small residual asymmetry (W_L vs W_L^{-1} , index order) is the field-level fingerprint of a transforming in the conjugate representation (Lecture 6, [p. 8***]).

11 Pure translations: plane waves [p. 23***]

For space-time translations, $\Lambda = \mathbb{I}$.

Equation for v^A :

$$v^A(m, \vec{p}, \sigma, x + a) = v^A(m, \vec{p}, \sigma, x) e^{ia \cdot p}$$

Equation for u^A :

$$u^A(m, \vec{p}, \sigma, x + a) = u^A(m, \vec{p}, \sigma, x) e^{-ia \cdot p}$$

These equations admit the following solutions:

$$v^A(m, \vec{p}, \sigma, x) = v^A(m, \vec{p}, \sigma) e^{ip \cdot x}, \quad u^A(m, \vec{p}, \sigma, x) = u^A(m, \vec{p}, \sigma) e^{-ip \cdot x}.$$

Obs. These are plane waves: eigenstates of momentum with eigenvalue p^μ . In other words, these coefficient functions — at least for the moment, since the translations are considered — share the same transformation properties of the physical states $|m, \vec{p}, \sigma\rangle$ they intend to describe.

We insert the above solutions into the equations for u^A, v^A .

12 The final form: fields and x -independent conditions [p. 24***–25***]

We write, for the v equation,

$$D(\Lambda^{-1})^A{}_B \underbrace{v^B(m, \vec{p}_\Lambda, \sigma, \Lambda x + a)} = \sum_{\sigma'} \underbrace{v^A(m, \vec{p}, \sigma', x)} e^{ia \cdot p_\Lambda} (D_s(W_L(\Lambda, p)))_{\sigma\sigma'}.$$

On the left,

$$v^B(m, \vec{p}_\Lambda, \sigma) e^{ip_\Lambda \cdot (\Lambda x + a)} = v^B(m, \vec{p}_\Lambda, \sigma) e^{ip_\Lambda \cdot \Lambda x} e^{ip_\Lambda \cdot a} = v^B(m, \vec{p}_\Lambda, \sigma) e^{ip \cdot x} e^{ip_\Lambda \cdot a},$$

since $p_\Lambda \cdot \Lambda x$ is a scalar product in Minkowski, and we used Lorentz-invariance: $\Lambda x \cdot p_\Lambda = x \cdot p$. On the right,

$$v^A(m, \vec{p}, \sigma') e^{ip \cdot x} e^{ia \cdot p_\Lambda}.$$

The terms $e^{ip \cdot x} e^{ia \cdot p_\Lambda}$ simplify. The previous equation becomes

$$D(\Lambda^{-1})^A{}_B v^B(m, \vec{p}_\Lambda, \sigma) = \sum_{\sigma'} v^A(m, \vec{p}, \sigma') [D_s(W_L(\Lambda, p))]_{\sigma\sigma'}$$

with a similar manipulation for the equation for u^A . We write

$$\begin{aligned} \Phi_-^A(x) &= \sum_\sigma \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} v^A(m, \vec{p}, \sigma) e^{ip \cdot x} a^\dagger(m, \vec{p}, \sigma) \\ D(\Lambda^{-1})^A{}_B v^B(m, \vec{p}_\Lambda, \sigma) &= \sum_{\sigma'} v^A(m, \vec{p}, \sigma') [D_s(W_L(\Lambda, p))]_{\sigma\sigma'} \end{aligned}$$

and, in similar fashion, [p. 25***]

$$\begin{aligned} \Phi_+^A(x) &= \sum_\sigma \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} u^A(m, \vec{p}, \sigma) e^{-ix \cdot p} a(m, \vec{p}, \sigma) \\ D(\Lambda^{-1})^A{}_B u^B(m, \vec{p}_\Lambda, \sigma) &= \sum_{\sigma'} u^A(m, \vec{p}, \sigma') [D_s(W_L(\Lambda, p)^{-1})]_{\sigma'\sigma} \end{aligned}$$

Comments.

- 1) The coefficient functions u^A and v^A cannot be anything but must verify the previous equations.
- 2) The fields inherit the same commutation/anticommutation relations of the creation and annihilation operators with which they are built.

▼ As a consequence, one finds immediately that

$$[\Phi_+^A(x), \Phi_-^B(y)]_\pm = \sum_\sigma \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p} u^A(m, \vec{p}, \sigma) v^B(m, \vec{p}, \sigma) e^{-ip \cdot (x-y)},$$

which does not vanish for $(x - y)^2 < 0$.

[This problem cannot be solved by making the observables out of creation or annihilation fields alone since they could not be Hermitian.]

The only way out is to combine $\Phi_+^A(x)$ and $\Phi_-^A(x)$ into a single object: [p. 26***]

$$\Phi^A(x) \equiv \kappa \Phi_+^A(x) + \eta \Phi_-^A(x)$$

with κ and η adjusted in such a way that

$$[\Phi^A(x), \Phi^B(y)]_{\pm} = [\Phi^A(x), \Phi^{\dagger B}(y)]_{\pm} = 0 \quad \text{if } (x - y)^2 < 0.$$

Added explanation (not in the original notes)

This is the crux of the whole construction. Let us spell out the impasse and the escape. *The impasse:* the elementary quanta demand *complex* combinations (a with $e^{-ip \cdot x}$, $a^\dagger$ with $e^{+ip \cdot x}$), but neither piece alone can enter observables (not Hermitian: $\Phi_+^\dagger \sim \Phi_-$), while their mixed bracket $[\Phi_+, \Phi_-]_{\pm}$ is an integral of $e^{-ip \cdot (x-y)}$ over the mass shell with *positive energies only* — and a purely positive-frequency function of $(x - y)$ cannot vanish on the open set of space-like separations (for the scalar case it is the standard function $\Delta_+(x - y)$, which decays but does not vanish outside the light cone). *The escape:* in the combination $\kappa \Phi_+ + \eta \Phi_-$, the spacelike bracket receives *two* contributions — one $\propto \kappa \eta$ from $[\Phi_+, \Phi_-]_{\pm}$ and one $\propto \eta \kappa$ from $[\Phi_-, \Phi_+]_{\pm}$ — which enter with a relative sign that depends on the choice of commutator vs anticommutator. Causality holds iff the two pieces cancel, and (as presumably shown in the next lecture) this cancellation works out only when integer-spin fields are quantized with commutators and half-integer-spin fields with anticommutators, with matched $|\kappa| = |\eta|$: the *spin-statistics theorem*, converting the “input” rule of Lecture 6 into a consequence of causality + relativity. The same cancellation requirement is also what ties a -type and $a^\dagger$ -type quanta together into one field — for complex fields, this forces the $a^\dagger$ part to create *antiparticles*: distinct particles with the same mass and spin, another celebrated corollary of the framework.

3) The equations for u^A and v^A can be written in a slightly different but equivalent form.

■ Equation for u^A . Starting from

$$D(\Lambda^{-1})^A{}_B u^B(m, \vec{p}_\Lambda, \sigma) = \sum_{\sigma'} u^A(m, \vec{p}, \sigma') [D_s(W_L^{-1})]_{\sigma' \sigma},$$

since D is a representation, $D(\Lambda^{-1}) = D(\Lambda)^{-1}$ (and likewise $D_s(W_L^{-1}) = D_s(W_L)^{-1}$):

$$[D(\Lambda)^{-1}]^A{}_B u^B(m, \vec{p}_\Lambda, \sigma) = \sum_{\sigma'} u^A(m, \vec{p}, \sigma') [D_s(W_L)^{-1}]_{\sigma' \sigma}.$$

We invert, multiplying by $D_s(W_L)$ and $D(\Lambda)$ both sides. We find

$$D(\Lambda)^A{}_B u^B(m, \vec{p}, \sigma) = \sum_{\sigma'} u^A(m, \vec{p}_\Lambda, \sigma') [D_s(W_L(\Lambda, p))]_{\sigma' \sigma}$$

Added explanation (not in the original notes)

Checking the inversion, and what the final equation says. Write the previous relation in matrix form (suppressing m): $D(\Lambda)^{-1} u(\vec{p}_\Lambda) = u(\vec{p}) D_s(W_L)^{-1}$, where $u(\vec{p})$ is the rectangular matrix with entries $u^A(\vec{p}, \sigma)$. Multiplying by $D(\Lambda)$ on the left gives $u(\vec{p}_\Lambda) = D(\Lambda) u(\vec{p}) D_s(W_L)^{-1}$; multiplying then by $D_s(W_L)$ on the right, $D(\Lambda) u(\vec{p}) = u(\vec{p}_\Lambda) D_s(W_L)$, which in components is exactly the boxed result. Read as a statement about representation theory, it says the collection $\{u^A(\vec{p}, \sigma)\}$ *intertwines* the finite-dimensional (generally non-unitary) representation $D(\Lambda)$ acting on the Lorentz index A with the unitary little-group representation D_s acting on the particle label σ : applying a Lorentz transformation to the field index is the same as boosting the momentum and Wigner-rotating the spin. Solving this intertwining condition for each choice of D — scalar, spinor, vector, ... — produces the familiar polarization objects of QFT (for the scalar: $u = v = 1$; for spin $\frac{1}{2}$: the Dirac $u_\sigma(p), v_\sigma(p)$ spinors; for spin 1: the polarization vectors $\varepsilon_\sigma^\mu(p)$), and is presumably the next step of the course.

(End of Lecture 7 — the notes continue with the solution of the consistency conditions and the construction of the causal field $\Phi^A(x)$ for specific spins.)